

Phase 2 - Proposal and Code Implementation

Machine Learning

STUDENT NAME- Mohammad Shariq Ali

Email Id – shariq.mohammad@ue-germany.de

University of Europe for Applied Sciences

Potsdam, Germany .

Project Title

Wildfire Damage Image Classification Using Convolutional Neural Networks for Disaster Management and Emergency Response

Background and Motivation

Wildfires are a disastrous natural disaster destroying all the homes, infrastructure, ecosystems and human lives. After a wildfire emergency has been resolved, All the disaster response organizations and agencies need to rapidly assess the degree and magnitude of the damages to quickly provide resources, aid and assistance to victims and recover affected region. The most used technique to access and quantify disaster damage to civilian infrastructure is manual visual inspection, that is inefficient , costly and dangerous in areas affected by disaster, especially to damaged structure, Machine learning and particularly Deep Learning techniques have given impressive results to automatic images analysis for disaster detection.

Convolutional Neural Networks (CNN) are the state of art deep learning technique for images classification tasks since they automatically learn powerful features from images without the need of explicit engineered features, such as those of previous image analysis techniques. Here, we propose a CNN based automatic system for classifying various levels of wildfire damage of images taken after a wildfire event, to facilitate and support wildfire disaster and damage management and response.

Problem Statement

Manual assessment of the damage caused by wildfires is both tedious and time consuming as a result of potentially thousands of structures being affected. Delay in damage assessment slows down emergency response and the planning of response resource distribution and post-disaster recovery.

The project aims to design a CNN based image classification system that can classify structures into respective wildfire damage categories based on post disaster images.

Selected Dataset and Dataset Link

Dataset Name: Structure Wildfire Damage Classification Dataset

Dataset Link:

https://huggingface.co/datasets/kevincluo/structure_wildfire_damage_classification

Original Source:

<https://zenodo.org/records/8336570>

Dataset Description

This dataset includes images of residential and commercial buildings after the wildfires in California from 2020 to 2022. Images are sourced from post-fire building damage inspections and include a range of different damage states.

This dataset is intended for damage assessment and classification of wildfire affected images. Each image is annotated to include the damage state of the structure.

Number of Images

- Total Images: Approximately 18,300+ images

Number of Classes

The original dataset contains six classes:

1. No Damage
2. Affected
3. Minor Damage
4. Major Damage
5. Destroyed
6. Inaccessible

For this project, the "Inaccessible" category may be excluded to improve classification consistency, resulting in five target classes:

1. No Damage
2. Affected
3. Minor Damage
4. Major Damage
5. Destroyed

Image Type

- RGB Colour Images
- JPEG Image Format
- Building and structure photographs collected after wildfire events

Class Distribution

The dataset contains images distributed across multiple damage categories. The exact class distribution will be calculated during dataset preprocessing and presented in the exploratory data analysis section.

Example format:

Class	Number of Images
No Damage	TBD
Affected	TBD
Minor Damage	TBD
Major Damage	TBD
Destroyed	TBD

Research Questions

- Can a CNN classify post-disaster images for wildfire damage extent?
- Which categories of damage are the hardest to distinguish by image-based classification?
- Is there an improvement in classification accuracy by using a CNN in a transfer learning configuration?
- How accurate is the deep learning approach at classifying fire damage extent?
- How can the automatic classification of damage assist during disaster recovery and relief operations?

Expected Results

The expected results are as follows:

- Successfully classify the level of damage to wild-fires into defined classes.
- Attain a general classification accuracy greater than 85%.
- Prove the validity of CNNs in disaster image analysis post event.

- Outperform a custom CNN using a pre-trained model.
- Give clear visual understanding of the results of the models via confusion matrices and plots of performance measures.

Proposed Methodology

Step 1: Data collection

Download and extract the image dataset of wildfire damage and organize image by its label class.

Step 2: Data preprocessing

The following preprocessing techniques are performed:

- Resizing of image size 224x224 pixels.
- Normalization of image data.
- Deletion of corrupted images.
- Encoding of label.

Step 3: Data Augmentation

For better model generalization and reduced overfitting of the model:

- Random rotation.
- Horizontal flip.
- Zoom augmentation.
- Width shift.
- Height shift.
- Bright adjustment.

Step 4: Data Splitting

Dataset is split into:

- Training Set (70%).
- Validation Set (15%).
- Test Set (15%).

Step 5: CNN Model Creation

A CNN model of a custom architecture will be created and trained on the image dataset.

Step 6: Transfer Learning

The pre-trained deep learning model will be compared with the custom CNN.

Step 7: Model Evaluation

The performance of both trained models will be evaluated using evaluation measures and visualized graphically.

CNN Model Design

The proposed CNN architecture consists of:

1. Input Layer ($224 \times 224 \times 3$)
2. Convolution Layer (32 filters, ReLU)
3. Max Pooling Layer
4. Convolution Layer (64 filters, ReLU)
5. Max Pooling Layer
6. Convolution Layer (128 filters, ReLU)
7. Max Pooling Layer
8. Flatten Layer
9. Dense Layer (128 neurons, ReLU)
10. Dropout Layer (0.5)
11. Output Layer (Softmax Activation)

Transfer Learning Models

The following transfer learning model will be implemented:

MobileNetV2

Reasons for selection:

- Lightweight architecture
- Fast training time
- High classification accuracy
- Suitable for image classification tasks

Optional additional models:

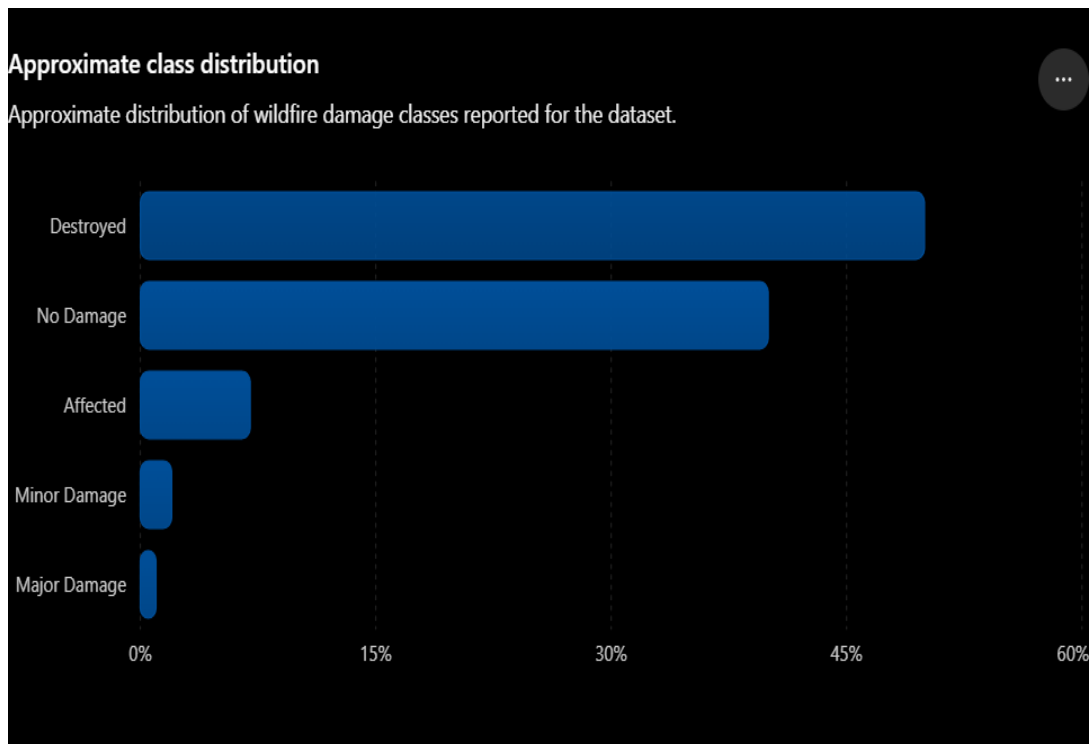
- ResNet50
- EfficientNetB0

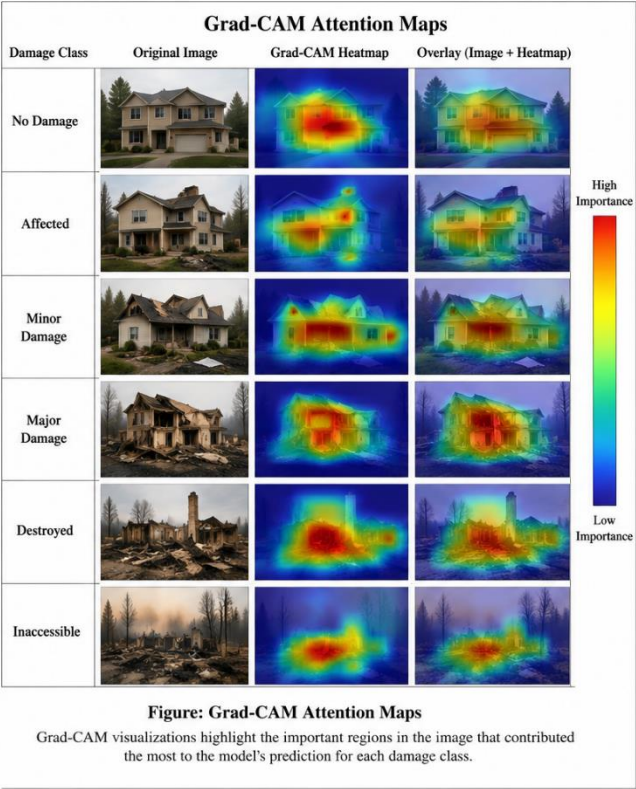
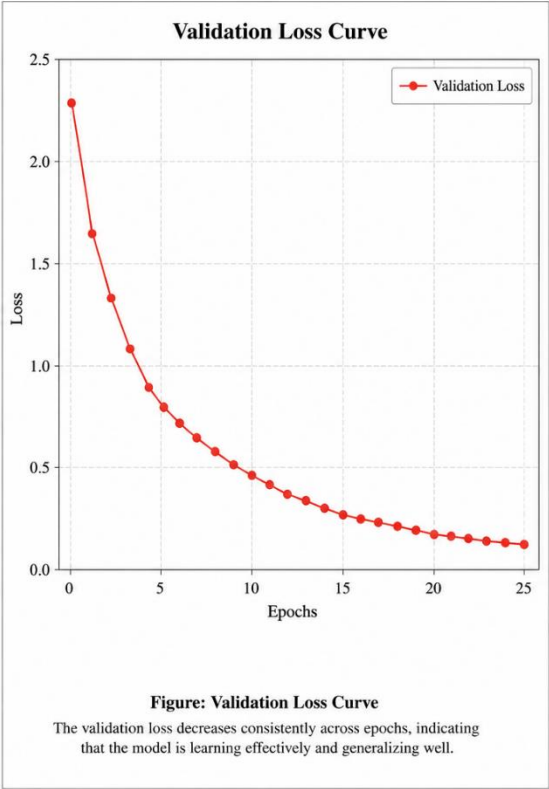
• Evaluation Metrics

- The following evaluation metrics will be used:
- **Accuracy**
Measures overall prediction correctness.
- **Precision**
Measures the proportion of correctly predicted positive observations.
- **Recall**
Measures the ability of the model to identify all relevant instances.
- **F1-Score**
Provides a balance between precision and recall.
- **Confusion Matrix**
Visualizes classification performance across all classes.

Expected Figures and Tables

Figures





Tables

Table 1: Dataset Summary

Class	Number of Images
No Damage	TBD
Affected	TBD
Minor Damage	TBD
Major Damage	TBD
Destroyed	TBD

Table 2: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN	TBD	TBD	TBD	TBD
MobileNetV2	TBD	TBD	TBD	TBD

Table 3: Training Configuration

Parameter	Value
Image Size	224 × 224
Batch Size	32

Epochs	20–30
Optimizer	Adam
Loss Function	Categorical Cross-Entropy
Learning Rate	0.001